



1.

(a)

迪摩根 $(x+y+z)' = x'y'z'$

x	y	z	$x+y+z$	$(x+y+z)'$	x	y	z	x'	y'	z'	$x'y'z'$
0	0	0	0	1	0	0	0	1	1	1	1
0	0	1	1	0	0	0	1	1	1	0	0
0	1	0	1	0	0	1	0	1	0	1	0
0	1	1	1	0	0	1	1	1	0	0	0
1	0	0	1	0	1	0	0	0	1	1	0
1	0	1	1	0	1	0	1	0	1	0	0
1	1	0	1	0	1	1	0	0	0	1	0
1	1	1	1	0	1	1	1	0	0	0	0

迪摩根 $(xyz)' = x' + y' + z'$

x	y	z	xyz	$(xyz)'$	x	y	z	x'	y'	z'	$x' + y' + z'$
0	0	0	0	1	0	0	0	1	1	1	1
0	0	1	0	1	0	0	1	1	1	0	1
0	1	0	0	1	0	1	0	1	0	1	1
0	1	1	0	1	0	1	1	1	0	0	1
1	0	0	0	1	1	0	0	0	1	1	1
1	0	1	0	1	1	0	1	0	1	0	1
1	1	0	0	1	1	1	0	0	0	1	1
1	1	1	1	0	1	1	1	0	0	0	0

(b)

分配律 $x+yz = (x+y)(x+z)$

[illegible]

(c) 分配率 $x(y+z) = xy + xz$

x	y	z	$y+z$	$x(y+z)$	x	y	z	xy	xz	$xy+xz$
0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	1	0	0	0
0	1	0	1	0	0	1	0	0	0	0
0	1	1	2	0	0	1	1	0	0	0
1	0	0	0	0	1	0	0	0	0	0
1	0	1	1	1	1	0	1	0	1	1
1	1	0	1	1	1	1	0	1	0	1
1	1	1	2	2	1	1	1	1	1	2

(d) 結合律: $x + (y + z) = (x + y) + z$

x	y	z	(y+z)	x+(y+z)		x	y	z	(x+y)	(x+y)+z
0	0	0	0	0		0	0	0	0	0
0	0	1	1	1		0	0	1	0	1
0	1	0	1	1		0	1	0	1	1
0	1	1	1	1		0	1	1	1	1
1	0	0	0	1		1	0	0	1	1
1	0	1	1	1		1	0	1	1	1
1	1	0	1	1		1	1	0	1	1
1	1	1	1	1		1	1	1	1	1

$$(e) x(yz) = (xy)z$$

x	y	z	yz	x(yz)
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

x	y	z	xy	(xy)z
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	1	0
1	1	1	1	1

2.

$$(a) \checkmark xy + xy' \\ = x(y + y') \\ = x \cdot 1 = x \neq$$

$$(b) \checkmark (x+y)(x+y') \\ = x \cdot x + x \cdot y' + y \cdot x + y \cdot y' \\ = x + x(y' + y) + 0 \\ = x + x \cdot 1 = x \neq$$

$$(c) \checkmark xyz + x'y + xy'z' \\ = xy(z + z') + x'y \\ = xy + x'y \\ = y(x + x') = y \neq$$

$$(d) \checkmark (x+y)'(x'+y')' \\ = (x'y')(xy) \rightarrow \text{using DeMorgan's The. to Extend.} \\ = x \cdot x' \cdot y \cdot y' \\ = 0 \neq$$

$$(e) (a+b+c')(a'b'+c) \\ = aa'b' + ac + a'bb' + bc + a'b'c' + c'c \\ = 0 + ac + 0 + bc + a'b'c' + 0 \\ = c(a+b) + a'b'c' \neq$$

$$(f) a'bc + abc' + abc + a'bc' \\ = b(a'c + ac' + ac + a'c') \\ = b[a(c' + c) + a'(c' + c)] \\ = b \neq$$

3.

$$\begin{aligned}
 (a) & \checkmark x y z + x' y + x y z' \\
 &= x y (z + z') + x' y \\
 &= x y + x' y = y \quad \text{✗}
 \end{aligned}$$

$$\begin{aligned}
 (b) & \checkmark x' y z + x z \\
 &= z (x' y + x) \xrightarrow{\text{消去律}} \\
 &= z (x + y) \quad \begin{matrix} a+a'b \\ = a+b \end{matrix} \quad \text{✗}
 \end{aligned}$$

$$\begin{aligned}
 (c) & \checkmark (x+y)' (x'+y') \\
 &= (x' y') (x' + y') \\
 &= x' \cdot x' y' + y' \cdot x' y' \\
 &= x' y' + x' y' = x' y' \quad \text{✗}
 \end{aligned}$$

$$\begin{aligned}
 (d) & \checkmark x y + x (w z + w z') \\
 &= x y + x w = x (y + w) \quad \text{✗}
 \end{aligned}$$

$$\begin{aligned}
 (e) & (y z' + x' w) (x y' + z w') \\
 &= \underline{y z'} \cdot \underline{x y'} + \underline{y z'} \cdot \underline{z w'} + \underline{x' w} \cdot \underline{x y'} + \underline{x' w} \cdot \underline{z w'} \\
 &= 0 \quad \text{✗}
 \end{aligned}$$

$$\begin{aligned}
 (f) & (x' + z') (x + y' + z') \\
 &= x' y' + x' z' + x z' + y' z' + z' \\
 &= z' (x + x') + z' (1 + y) + x' y' \\
 &= z' + z' + x' y' = z' + x' y' \quad \text{✗}
 \end{aligned}$$

4.

$$\begin{aligned}
 (a) & \checkmark x' y' z' + y + x y' z' \\
 &= y' z' (x' + x) + y \\
 &= y + y' z' \quad \leftarrow \text{消去律!} \\
 &= y + z' \quad \underline{a+a'b=a+b} \quad \text{✗}
 \end{aligned}$$

$$\begin{aligned}
 (b) & x' y (x' + z') + x' y + x y z \\
 &= x' y + x' y z' + x' y + x y z \\
 &= x' y + x' y z' + x y z \\
 &= x' y + x y z \quad \leftarrow \begin{matrix} \text{吸收律} \\ a+ab=a \end{matrix} \\
 &= y (x' + x z) \quad \leftarrow x' + x z = x' + z, \text{ PPT P. 26} \\
 &= y (x' + z) \quad \text{✗}
 \end{aligned}$$

$$\begin{aligned}
 (c) & (x + y z)' + (x + y' z')' \\
 &= x' (y z)' + x' (y' z')' \\
 &= x' \cdot (y' + z') + x' \cdot (y + z) \\
 &= x' \cdot (y' + y + z' + z) = x' \quad \text{✗}
 \end{aligned}$$

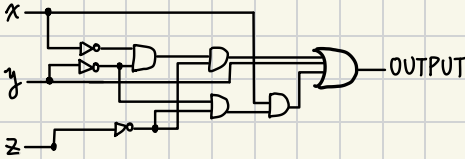
$$\begin{aligned}
 (d) & (w' + x) (w + y) (x' + y) (w + x y z) \\
 &= (w x' + w y + \underline{y x' + y}) (w' + x) (w + x y z) \\
 &= (w x' + w y + \underline{y}) (w' + x) (w + x y z) \\
 &= (w x' + y) (w' + x) (w + x y z) \\
 &= (\cancel{w' w x'} + \cancel{x w x} + y w' + x y) (w + x y z) \\
 &= y (w' + x) (w + x y z) = y (x y z w' + x w + x y z) \\
 &= x y (\underline{y z w} + y z + w) \\
 &= x y (y z + w) = x y (z + w)
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad & wx'y'z' + wy' + wx'y'z' \\
 &= w(xy'z' + x'y'z' + y') \\
 &= w(y'z' + y') = wy'
 \end{aligned}$$

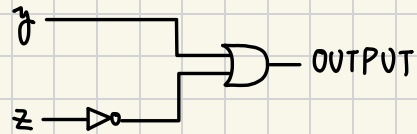
7.

(a)

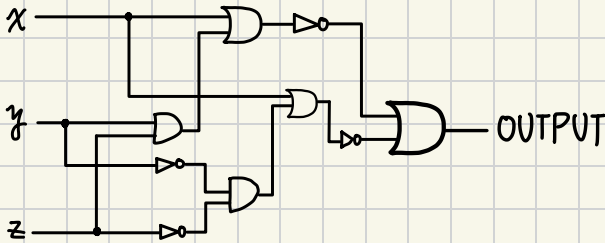
$$1. x'y'z' + y + xy'z'$$



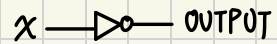
$$2. y + z'$$



$$(c) \quad 1. (x + yz)' + (x + y'z')'$$

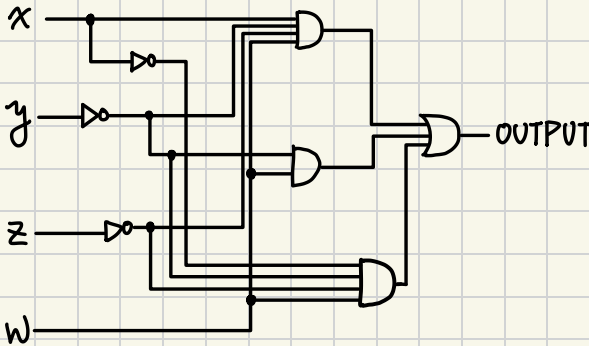


$$2. x'$$

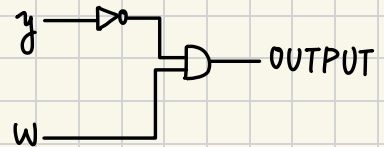


(e)

$$1. wx'y'z' + wy' + wx'y'z'$$



$$2. wy'$$



8.

$$F = x'y + yz'$$

$$\begin{aligned} F' &= (x'y + yz')' = (x'y)'(yz')' \\ &= (x+y')(y'+z) \\ &= xy' + xz + \underline{y'y'z} \\ &= xy' + \underline{y} + xz = y' + xz \end{aligned}$$

$$F \cdot F' = (x'y + yz') \cdot (y' + xz)$$

$$= (0 + 0 + 0 + 0) = 0$$

$$F + F' = (x'y + yz') + (y' + xz)$$

$$-y = 0:$$

$$(x' \cdot 0 + 0 \cdot z') + (1 + xz) = 1 + xz = 1$$

$$-y = 1:$$

$$(x' \cdot 1 + 1 \cdot z') + (0 + xz) = x' + z' + xz$$

$$= \underline{x' + xz} + z'$$

$$= \underline{x' + z} + z' = x' + 1 = 1$$

$\Rightarrow F + F'$ 恒為 1, 得證。

9.

$$(a) xyz + x'y'z'$$

$$(xyz + x'y'z')$$

$$= (xyz)'(x'y'z')'$$

$$= (x' + y' + z')(x + y + z) \neq$$

$$(b) (x+y')(y+z')(z+x')$$

$$\Rightarrow [(x+y')(y+z')(z+x')]'$$

$$= (x+y')'(y+z')'(z+x')'$$

$$= (x'y)(y'z)(z'x)$$

$$(c) w'x' + w(x+y+z)$$

$$[(w'x') + w(x+y+z)]'$$

$$= (w+x)(w' + (x+y+z)')$$

$$= (w+x)(w' + (x'y'z'))$$

$$= 0 + w \cdot x'y'z' + xw' + 0$$

$$= wx'y'z' + w'x \neq$$

10.

$$\text{let } F_1 = \sum_{i \in S_1} m_i, F_2 = \sum_{j \in S_2} m_j$$

m_i, m_j 表第 i 或 j 个最小項, S_1, S_2 表使 F_1, F_2 取值為 1 的指數集合

(a) $E = F_1 + F_2$ 包含 F_1 和 F_2 的最小項之和

$$F_1 + F_2 = \sum_{i \in S_1} m_i + \sum_{j \in S_2} m_j = \sum_{k \in (S_1 \cup S_2)} m_k$$

(b) $G = F_1 \cdot F_2$ 只包含 F_1 和 F_2 共有的 Minterms

$$F_1 \cdot F_2 = \sum_{i \in S_1} m_i \cdot \sum_{j \in S_2} m_j = \sum_{k \in S_1 \cap S_2} m_k$$

11.

(a) $F = x'z' + xy + yz$

x	y	z	x'	z'	x'z'	xy	yz	x'z' + xy + yz	
0	0	0	1	1	1	0	0	1	(m ₀)
0	0	1	1	0	0	0	0	0	
0	1	0	1	1	1	0	0	1	(m ₂)
0	1	1	1	0	0	0	1	1	(m ₃)
1	0	0	0	1	0	0	0	0	
1	0	1	0	0	0	0	0	0	
1	1	0	0	1	0	1	0	1	(m ₆)
1	1	1	0	0	0	1	1	1	(m ₇)

$F(x,y,z) = \sum(0,2,3,6,7)$

(b) $F = a'b'c' + a'bc + ab'c' + abc$

a	b	c	a'	b'	c'	a'b'c'	a'bc	ab'c'	abc	a'b'c' + a'bc + ab'c' + abc
0	0	0	1	1	1	1	0	0	0	1 (m ₀)
0	0	1	1	1	0	0	0	0	0	0
0	1	0	1	0	1	0	0	0	0	0
0	1	1	1	0	0	0	1	0	0	1 (m ₃)
1	0	0	0	1	1	0	0	1	0	1 (m ₄)
1	0	1	0	1	0	0	0	0	0	0
1	1	0	0	0	1	0	0	0	0	0
1	1	1	0	0	0	0	0	0	1	1 (m ₇)

$$F(a,b,c) = \sum(0,3,4,7)$$

12.

$A = 1100 \ 1010$

$B = 1001 \ 0011$

(a) AND

$$\begin{array}{r} 1100 \ 1010 \\ \text{AND } 1001 \ 0011 \\ \hline 1000 \ 0010 \end{array}$$

(b) OR

$$\begin{array}{r} 1100 \ 1010 \\ \text{OR } 1001 \ 0011 \\ \hline 1101 \ 1011 \end{array}$$

(c) XOR

$$\begin{array}{r} 1100 \ 1010 \\ \text{XOR } 1001 \ 0011 \\ \hline 0101 \ 1001 \end{array}$$

(d) NOT A

$$\begin{array}{r} \text{NOT } 1100 \ 1010 \\ \hline 0011 \ 0101 \end{array}$$

(e) NOT B

$$\begin{array}{r} \text{NOT } 1001 \ 0011 \\ \hline 0110 \ 1100 \end{array}$$

14.

$$F = x'y + xy' + xz$$

(a) AND, OR, NOT $(x'y) + (xy') + (xz)$

(b) OR, NOT

<key: $ab = (a+b)' / a'b' = (a+b)'$ >

$$x'y = (x+y)', \quad xy' = (x'+y)', \quad xz = (x'+z)'$$

$$(x+y)' + (x'+y)' + (x'+z)'$$

(c) AND, NOT

<key: $a+b = (a'b)' / a'b' = (a \cdot b)'$ >

$$(x'y) + (xy') = [(x'y)'(xy')']'$$

$$[(x'y)'(xy')']' + (xz) = [(x'y)'(xy')'(xz)']'$$

(d) NAND, NOT

<key: $(a \uparrow b)' = (a \cdot b)' / (a \uparrow b) = a \cdot b$ >

$$x'y = x' \uparrow y, \quad xy' = x \uparrow y', \quad xz = x \uparrow z$$

$$[(x \uparrow y) \uparrow (x \uparrow y')] \uparrow (x \uparrow z)$$

(e) NOR, NOT

<key: $(a+b)' = a \downarrow b, \quad ab = (a'+b')'$ >

$$x'y = (x+y)' = x \downarrow y$$

$$xy' = (x'+y)' = x' \downarrow y$$

$$xz = (x'+z)' = x' \downarrow z$$

$$[(x \downarrow y) \downarrow (x' \downarrow y)] \downarrow (x' \downarrow z)$$

NAND Truth Table

x	y	$x \uparrow y$
0	0	1
0	1	1
1	0	1
1	1	0

NOR Truth Table

x	y	$x \downarrow y$
0	0	1
0	1	0
1	0	0
1	1	0

15.

A	B	C	T ₁	T ₂
0	0	0	0	1✓
0	0	1	✓1	0
0	1	0	✓1	1✓
0	1	1	✓1	0
1	0	0	0	0
1	0	1	✓1	1✓
1	1	0	0	0
1	1	1	✓1	1✓

T₁:

A \ BC	00	01	10
0	0	1	1
1	0	1	0

不論B為何
均有C=1
⇒ C

A=0 ⇒ A'
B=1 ⇒ A'B
⇒ A'B

$$T_1 = (A'B) + C$$

T₂:

A \ BC	00	01	10
0	1	1	0
1	0	0	1

BC 00
BC 10
都有C=0, A=0
⇒ A'C'

BC 01
BC 11
都有A=1, C=1
⇒ AC

$$T_2 = A'C' + AC$$

17.

(a) $(ac+b)(ab+d)$

a	b	c	d	ac	ac+b	ab	ab+d	(ac+b)(ab+d)
0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	1
0	0	1	0	0	0	0	0	2
0	0	1	1	0	0	0	1	3
0	1	0	0	0	1	0	0	4
0	1	0	1	0	1	0	1	5
0	1	1	0	0	1	0	0	6
0	1	1	1	0	1	0	1	7
1	0	0	0	0	0	0	0	8
1	0	0	1	0	0	0	1	9
1	0	1	0	1	1	0	0	10
1	0	1	1	1	1	0	1	11
1	1	0	0	0	1	1	1	12
1	1	0	1	0	1	1	1	13
1	1	1	0	1	1	1	1	14
1	1	1	1	1	1	1	1	15

$$\Sigma(5, 7, 11, 12, 13, 14, 15) = \Pi(0, 1, 2, 3, 4, 6, 8, 9, 10)$$

$$18. F = w'xy + w'y z + wy'z + w'y'z + xy$$

$$(c) \underline{w'xy + xy} + w'y z + wy'z + w'y'z$$

$$= xy + w'y z + w'y'z + wy'z$$

$$= xy + w'z(y + y') + wy'z$$

$$= xy + w'z + wy'z$$

$$= xy + z(w' + wy') \rightarrow \text{消去律: } a + a'b = a + b$$

$$= xy + z(w' + y')$$

$$19. F(A, B, C, D) = AB' + BC + ACD'$$

AB'項

$$A=1, B=0$$

$$A=1, B=0, C=0, D=0 \quad 1000_2 = 8$$

$$A=1, B=0, C=0, D=1 \quad 1001_2 = 9$$

$$A=1, B=0, C=1, D=0 \quad 1010_2 = 10$$

$$A=1, B=0, C=1, D=1 \quad 1011_2 = 11$$

Ans:

BC項

$$B=1, C=1$$

$$A=0, B=1, C=1, D=0 \quad 0110_2 = 6$$

$$A=0, B=1, C=1, D=1 \quad 0111_2 = 7$$

$$A=1, B=1, C=1, D=0 \quad 1110_2 = 14$$

$$A=1, B=1, C=1, D=1 \quad 1111_2 = 15$$

$$\Sigma(6, 7, 8, 9, 10, 11, 14, 15)$$

$$= \Pi(0, 1, 2, 3, 4, 5, 12, 13)$$

ACD'

$$A=1, B=0, C=1, D=0 \quad 1010_2 = 10_x$$

$$A=1, B=1, C=1, D=0 \quad 1110_2 = 14_x$$

20.

$$(a) F(w, x, y, z) = \sum (1, 5, 7, 11, 12, 14, 15)$$

$$F'(w, x, y, z) = \sum (0, 2, 3, 4, 6, 8, 9, 10, 13)$$

$$(b) F(x, y, z) = \pi(2, 4, 5)$$

$$F'(x, y, z) = \pi(0, 1, 3, 6, 7)$$

21. $F(x, y, z) = \sum (2, 3, 5, 7)$

$$= \pi(0, 1, 4, 6)$$

$$F(A, B, C, D) = \pi(1, 2, 3, 5, 8, 13)$$

$$= \sum (0, 4, 6, 7, 9, 10, 11, 12, 14, 15)$$

22.

$$(a) (w + xy')(x + y'z)$$

$$= wx + wy'z + \underline{xy' + xy'z}$$

$$= \underline{wx} + \underline{wy'z} + \underline{xy'}$$

$$= (w + xy')(x + y'z)$$

$$= (w + x)(w + y')(x + y'z)$$

$$\Rightarrow \sum (4, 5, 9, 12, 13, 14, 15)$$

$$\pi(0, 1, 2, 3, 6, 7, 8, 10, 11)$$

$$(b) xy + (w' + y'z')(z' + x'y')$$

$$= xy + w'z' + x'y'w' + \underline{y'z' + x'y'z'}$$

$$= xy + w'z' + x'y'w' + y'z'$$

1. wx

w	x	y	z	
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15

2. wy'z

w	x	y'	z	
1	0	0	1	9
1	1	0	1	\

3. xy'

w	x	y'	z	
0	1	0	0	4
0	1	0	1	5
1	1	0	0	\
1	1	0	1	\

